Linear Algebra and Its Applications

Section 1.10: Linear Models in Business, Science, and Engineering

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.10 Linear models in business, science, and engineering

A model begins by defining its variables and their units. Matrix columns then describe what one unit of each input contributes, and rows record one output requirement at a time. A mathematically consistent nutritional system is not physically feasible if it requires negative amounts of food. In a circuit, signs depend on the chosen loop directions and battery polarities. In a migration model, column j records where people starting in region j go, so its entries sum to 1 when nobody leaves the system. The labels [M] in the text permit matrix software; exact reduced matrices and substitution checks are supplied below rather than treating a software answer as a substitute for interpreting the model.

Practice problem 1.10.P1: Express the diet example as a matrix equation

Let x_1, x_2, x_3 be the amounts of nonfat milk, soy flour, and whey, respectively, measured in 100-gram units. Put the nutrient contributions per unit into columns and the targets into the right-hand side:

$$A = \begin{bmatrix} 36 & 51 & 13 \\ 52 & 34 & 74 \\ 0 & 7 & 1.1 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad b = \begin{bmatrix} 33 \\ 45 \\ 3 \end{bmatrix}, \qquad Ax = b.$$

The rows specify protein, carbohydrate, and fat in grams. For example, the first row says $36x_1 + 51x_2 + 13x_3 = 33$, so it adds the protein from all three ingredients. The actual food masses in grams are $100x_j$.

Exercise 1.10.1: A two-cereal mixture with four requirements

(a) Define the serving counts and the vector equation.

Let x be servings of Cheerios and y servings of Natural Cereal. With rows ordered as calories, protein, carbohydrate, and fat,

$$x \begin{bmatrix} 110 \\ 4 \\ 20 \\ 2 \end{bmatrix} + y \begin{bmatrix} 130 \\ 3 \\ 18 \\ 5 \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

(b) Solve and check all the requirements.

The equivalent matrix equation is

$$\begin{bmatrix} 110 & 130 \\ 4 & 3 \\ 20 & 18 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

Use the protein equation $4x + 3y = 9$ and fat equation $2x + 5y = 8$. Subtract twice the latter from the former: $-7y = -7$, hence $y = 1$. Then $2x + 5 = 8$ gives $x = 3/2$. Check the other rows: $110(3/2) + 130 = 295$ and $20(3/2) + 18 = 48$. Both serving counts are nonnegative, so the solution is physically meaningful.

Conclusion. Mix 1.5 servings of Cheerios with 1 serving of Natural Cereal.

Exercise 1.10.2: A specified mixture and a new nutritional target

(a) Build the contribution matrix.

Order the rows as calories, protein, fiber, and fat, and the columns as Shredded Wheat and Crispix. Then

$$B = \begin{bmatrix} 160 & 110 \\ 5 & 2 \\ 6 & 0.1 \\ 1 & 0.4 \end{bmatrix}, \quad u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad Bu = \begin{bmatrix} 700 \\ 19 \\ 18.2 \\ 3.8 \end{bmatrix}.$$

For instance the fiber amount is $6(3) + 0.1(2) = 18.2$ grams.

(b) Solve for the requested different mixture.

The desired equation is $B(x, y)^T = (130, 3.20, 2.46, 0.64)^T$. Its reduced augmented matrix is

$$\begin{bmatrix} 1 & 0 & 0.4 \\ 0 & 1 & 0.6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $x = 0.4$ and $y = 0.6$. To check all rows, the totals are $64 + 66 = 130$ calories, $2 + 1.2 = 3.20$ protein, $2.4 + 0.06 = 2.46$ fiber, and $0.4 + 0.24 = 0.64$ fat. These nonnegative proportions also add to one serving.

Conclusion. The target is attainable with 0.4 serving of Shredded Wheat and 0.6 serving of Crispix.

Exercise 1.10.3: Compare two versions of a lunch

Let x, y, z be servings of the pasta dish, broccoli, and chicken. Use calories, protein, and fiber as the three rows. Solving each system requires all three targets, not only the calories.

(a) Use the original Mac and Cheese.

$$\begin{bmatrix} 270 & 51 & 70 \\ 10 & 5.4 & 15 \\ 2 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Exact row reduction, treating the decimals as the stated rational numbers, gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 0 & 8205/8323 \\ 0 & 1 & 0 & 12850/8323 \\ 0 & 0 & 1 & 6550/8323 \end{bmatrix}.$$

Hence $(x, y, z) \approx (0.98582, 1.54391, 0.78698)$. Every variable is determined by a pivot, so the solution is unique. Substituting the exact fractions into the original three rows gives $(400, 30, 10)^T$. All three fractions are positive, so the mixture is feasible.

(b) Replace the pasta column by the whole-wheat-shells data.

Only that ingredient's column changes:

$$\begin{bmatrix} 260 & 51 & 70 \\ 9 & 5.4 & 15 \\ 5 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Reduction gives $(x, y, z) = (5470, 4400, 5180)/5023$, approximately $(1.08899, 0.87597, 1.03126)$. Substitution again gives exactly 400 calories, 30 grams protein, and 10 grams fiber before rounding. The broccoli portion drops from about 1.54 to 0.88 serving, in line with the stated preference.

Conclusion. To two decimal places the mixtures are $(0.99, 1.54, 0.79)$ and $(1.09, 0.88, 1.03)$ servings in the stated ingredient order. Use the exact fractions for exact nutritional totals.

Exercise 1.10.4: A consistent diet equation can be physically impossible

(a) Specify the four variables and four nutrient rows.

Let x_1, x_2, x_3, x_4 be 100-gram units of nonfat milk, soy flour, whey, and isolated soy protein. In the row order protein, carbohydrate, fat, calcium, the equation is

$$\begin{bmatrix} 36 & 51 & 13 & 80 \\ 52 & 34 & 74 & 0 \\ 0 & 7 & 1.1 & 3.4 \\ 1.26 & 0.19 & 0.8 & 0.18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 33 \\ 45 \\ 3 \\ 0.8 \end{bmatrix}.$$

Each coefficient is grams of that nutrient per 100 grams of ingredient.

(b) Solve and interpret the signs.

Exact reduction yields the unique solution

$$x = \begin{bmatrix} 2264687/3531104 \\ 1921341/3531104 \\ -326887/3531104 \\ -1468529/7062208 \end{bmatrix} \approx \begin{bmatrix} 0.64135 \\ 0.54412 \\ -0.09257 \\ -0.20794 \end{bmatrix}.$$

Multiplication of the exact vector by the original matrix reproduces all four targets, so the algebraic equations are consistent. But the whey and isolated-protein amounts are negative. One cannot subtract food already absent from the mixture, so this unique algebraic solution is not a permissible recipe. Because it is unique, there is no different nonnegative solution to these same equations.

Conclusion. These four ingredients cannot meet all four targets with nonnegative amounts under the supplied nutrient data.

Circuit convention for Exercises 5–8. Use the counterclockwise loop directions drawn in the textbook. For each loop, add all resistances on its boundary to get the diagonal coefficient. A shared resistor carries the difference of the adjacent loop currents, so its resistance enters the neighboring column with a minus sign. On the right, crossing a battery from its short plate to its long plate contributes a positive voltage; the reverse crossing contributes a negative voltage. These rules explain every sign in $Ri = v$. A negative current means the actual direction is opposite to the assumed counterclockwise direction.

Exercise 1.10.5: Derive the loop equations and solve them

The four diagonal resistance sums are $1+1+4+5=11$, $3+5+1+1=10$, $4+1+2+2=9$, and $1+2+3+4=10$. The successive shared resistances are $5, 1, 2$. Battery crossings give $20+30=50$, $-30-10=-40$, $10+20=30$, and $-20-10=-30$ volts.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 11 & -5 & 0 & 0 \\ -5 & 10 & -1 & 0 \\ 0 & -1 & 9 & -2 \\ 0 & 0 & -2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 50 \\ -40 \\ 30 \\ -30 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{265}{72} \\ -\frac{137}{72} \\ \frac{185}{72} \\ -\frac{179}{72} \end{bmatrix} \approx \begin{bmatrix} 3.6806 \\ -1.9028 \\ 2.5694 \\ -2.4861 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.6: Derive the loop equations and solve them

The diagonal sums are $3+2+1=6$, $1+4+4=9$, $4+1+2=7$, and $2+2+3=7$. Adjacent loops share resistances $1, 4, 2$. Every battery on the left is crossed from short plate to long plate in the chosen direction, giving voltages $30, 20, 40, 10$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 6 & -1 & 0 & 0 \\ -1 & 9 & -4 & 0 \\ 0 & -4 & 7 & -2 \\ 0 & 0 & -2 & 7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \\ 40 \\ 10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3630}{571} \\ \frac{4650}{571} \\ \frac{6700}{571} \\ \frac{2730}{571} \end{bmatrix} \approx \begin{bmatrix} 6.3573 \\ 8.1436 \\ 11.734 \\ 4.7811 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 30, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.7: Derive the loop equations and solve them

Keep the order I_1 upper left, I_2 lower left, I_3 lower right, I_4 upper right. The diagonal sums are $1+7+4=12$, $7+6+2=15$, $6+5+3=14$, and $4+5+4=13$. Shared resistances are 7 between loops 1 and 2, 6 between 2 and 3, 5 between 3 and 4, and 4 between 4 and 1. The source voltages are $40, 30, 20, -10$; the last is negative because loop 4 crosses that battery against its positive direction.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 12 & -7 & 0 & -4 \\ -7 & 15 & -6 & 0 \\ 0 & -6 & 14 & -5 \\ -4 & 0 & -5 & 13 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 40 \\ 30 \\ 20 \\ -10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{119910}{10487} \\ \frac{110640}{10487} \\ \frac{84270}{10487} \\ \frac{61240}{10487} \end{bmatrix} \approx \begin{bmatrix} 11.434 \\ 10.55 \\ 8.0357 \\ 5.8396 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 40, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.8: Derive the loop equations and solve them

The loop order is the four outer regions $1, 2, 3, 4$, then the central diamond 5. Their diagonal sums are $3+1+4+1=9$, $1+1+3+2=7$, $2+2+3+3=10$, $1+1+2+3=7$, and $4+3+3+2=12$. The central loop shares resistances $4, 3, 3, 2$ with the four outer loops. Additional shared stems are 1 between 1 and 4, 1 between 1 and 2, 2 between 2 and 3, and 3 between 3 and 4. Its own battery contribution is zero; the outer contributions are $50, -30, 20, -40$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 9 & -1 & 0 & -1 & -4 \\ -1 & 7 & -2 & 0 & -3 \\ 0 & -2 & 10 & -3 & -3 \\ -1 & 0 & -3 & 7 & -2 \\ -4 & -3 & -3 & -2 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 50 \\ -30 \\ 20 \\ -40 \\ 0 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3019}{755} \\ -\frac{3309}{755} \\ -\frac{683}{755} \\ -\frac{4382}{755} \\ -\frac{722}{755} \end{bmatrix} \approx \begin{bmatrix} 3.9987 \\ -4.3828 \\ -0.90464 \\ -5.804 \\ -0.95629 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Migration convention for Exercises 9–13. Put the current populations into a column vector. In each migration matrix, column j tells what fractions of starting population j arrive in the various destinations. Thus $x_{k+1} = Mx_k$ and two years later means $x_2 = M(Mx_0)$, not Mx_0.

Exercise 1.10.9: A seven-percent outward migration rate

Let $x_k = (c_k, s_k)^T$ denote city and suburban populations k years after 2010. The matrix and initial population are

$$M = \begin{bmatrix} 0.93 & 0.05 \\ 0.07 & 0.95 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 800000 \\ 500000 \end{bmatrix}, \qquad x_{k+1} = Mx_k.$$

For example $c_{k+1} = 0.93c_k + 0.05s_k$: city residents who stay plus suburban residents who move into the city. First multiplication gives $x_1 = (769000, 531000)^T$. Multiplying once more gives

$$x_2 = \begin{bmatrix} 741720 \\ 558280 \end{bmatrix}.$$

The total remains 1300000, as a check on the column sums and arithmetic.

Conclusion. In 2012 the model predicts 741720 city residents and 558280 suburban residents.

Exercise 1.10.10: A six-percent outward migration rate

With city first and suburbs second,

$$M = \begin{bmatrix} 0.94 & 0.04 \\ 0.06 & 0.96 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 10000000 \\ 800000 \end{bmatrix}, \quad x_{k+1} = Mx_k.$$

Here the first column says 94% of city residents stay and 6% move to the suburbs. The second says 4% move in and 96% stay in the suburbs. Successive multiplications give

$$x_1 = \begin{bmatrix} 9432000 \\ 1368000 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 8920800 \\ 1879200 \end{bmatrix}.$$

Both totals equal the initial 10800000.

Conclusion. The 2012 prediction is 8920800 in the city and 1879200 in the suburbs, subject to the model's exclusions.

Exercise 1.10.11: Estimate migration rates before iterating

(a) Divide each flow by its population of origin.

The fraction leaving California is $516100/31524000 \approx 0.0163716533$, hence 0.01637 to five decimal places. The fraction moving into California from elsewhere is $381262/228680000 \approx 0.0016672293$, hence 0.00167. Put California first. The rounded migration model is

$$M = \begin{bmatrix} 0.98363 & 0.00167 \\ 0.01637 & 0.99833 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 31524000 \\ 228680000 \end{bmatrix}.$$

The diagonal entries are the complements of the corresponding outward rates, not independently rounded unrelated quantities.

(b) Apply six annual steps to reach 2000.

Compute $x_{k+1} = Mx_k$ six times, retaining full arithmetic with the specified five-place coefficients. The result is

$$x_6 \approx \begin{bmatrix} 30754526.4814 \\ 229449473.5186 \end{bmatrix}.$$

To the nearest person, these are 30754526 in California and 229449474 elsewhere. Their sum is the unchanged total 260204000. The fractional entries before rounding are population estimates, not literal fractional people. Using unrounded migration rates would yield slightly different estimates, so it is important to state which rates were used.

Conclusion. The model predicts about 30.755 million in California and 229.449 million elsewhere, with births, deaths, and international migration excluded as required.

Exercise 1.10.12: Follow a fleet through two daily transitions

Use the order airport, east, west. The columns correspond to the rental origin and rows to the return location, so

$$M = \begin{bmatrix} 0.97 & 0.05 & 0.10 \\ 0 & 0.90 & 0.05 \\ 0.03 & 0.05 & 0.85 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 295 \\ 55 \\ 150 \end{bmatrix}.$$

Monday to Tuesday gives $x_1 = Mx_0 = (303.9, 57, 139.1)^T$. Tuesday to Wednesday gives

$$x_2 = Mx_1 = \begin{bmatrix} 311.543 \\ 58.255 \\ 130.202 \end{bmatrix}.$$

For example the first entry is $0.97(303.9) + 0.05(57) + 0.10(139.1)$. The three entries sum to 500 before rounding, matching the fleet size.

Conclusion. Approximately 312 cars at the airport, 58 at the east location, and 130 at the west location on Wednesday.

Exercise 1.10.13: Twenty iterations and their limiting pattern

The source model uses $M = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$. The total population is 1000000 in both parts. Therefore, if c_k is the city population, the suburb population is $1000000 - c_k$, and the first row of the recurrence becomes

$$c_{k+1} = 0.95c_k + 0.03(1000000 - c_k) = 0.92c_k + 30000.$$

Subtract 375000 from both sides to obtain $c_{k+1} - 375000 = 0.92(c_k - 375000)$. Repeated substitution gives

$$c_k = 375000 + (c_0 - 375000)(0.92)^k, \qquad s_k = 1000000 - c_k.$$

This formula both checks the matrix iterations and explains their pattern.

(a) Start from $(600000, 400000)^T$**.**

The city component is $375000 + 225000(0.92)^k$.

(b) Start from $(350000, 650000)^T$**.**

The city component is $375000 - 25000(0.92)^k$. The following table lists all twenty requested vectors, with entries rounded to the nearest person only for display.

	Part (a)		Part (b)	
k	City	Suburbs	City	Suburbs
1	582000	418000	352000	648000
2	565440	434560	353840	646160
3	550205	449795	355533	644467
4	536188	463812	357090	642910
5	523293	476707	358523	641477
6	511430	488570	359841	640159
7	500515	499485	361054	638946
8	490474	509526	362170	637830
9	481236	518764	363196	636804
10	472737	527263	364140	635860
11	464918	535082	365009	634991
12	457725	542275	365808	634192
13	451107	548893	366544	633456
14	445018	554982	367220	632780
15	439417	560583	367843	632157
16	434264	565736	368415	631585
17	429522	570478	368942	631058
18	425161	574839	369427	630573
19	421148	578852	369872	630128
20	417456	582544	370283	629717

Since $0 < 0.92 < 1$, its powers tend to zero. In (a) the city population decreases toward 375000, passing below the suburban population between years 7 and 8 (first below at the year-8 update). In (b) it increases toward the same 375000. The suburb populations approach 625000 from the opposite sides. The common limiting proportions come from the migration rule, not from the initial distribution.

Conclusion. Both initial distributions tend to $(375000, 625000)^T$, but they approach it from different sides.

Exercise 1.10.14: Temperature models obey superposition

(a) Write one averaging equation for each interior point.

Order the points as 1 upper left, 2 upper right, 3 lower right, 4 lower left, as in the diagrams. Moving

the two interior-neighbor temperatures to the left gives the common coefficient matrix

$$K = \begin{bmatrix} 4 & -1 & 0 & -1 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ -1 & 0 & -1 & 4 \end{bmatrix}.$$

For plate A, each point touches boundary temperatures 20 and 0, so $KT_A = (20, 20, 20, 20)^T$. Direct substitution gives $T_A = (10, 10, 10, 10)^T$. For plate B, the two relevant boundary values sum to $10, 40, 50, 20$ in point order. Thus $KT_B = (10, 40, 50, 20)^T$, and solving gives

$$T_B = (10, 17.5, 20, 12.5)^T.$$

For example its first equation is $4(10) - 17.5 - 12.5 = 10$, and its third is $4(20) - 17.5 - 12.5 = 50$. Their sum is

$$T_A + T_B = (20, 27.5, 30, 22.5)^T,$$

exactly the temperature list cited from Section 1.1. The boundary data there are the sums of these two plates' boundary data, so the same additivity appears at the interior points.

(b) Triple all of plate A's boundary temperatures.

The new right-hand side is $(60, 60, 60, 60)^T$, three times the old one. Hence $T = (30, 30, 30, 30)^T = 3T_A$. This is also checked directly: each row is $4(30) - 30 - 30 = 60$.

(c) Formulate and justify the general rule.

The map from the eight boundary values to the four interior values is linear: adding boundary data adds interior solutions, and scaling boundary data scales the interior solution. To justify uniqueness, if two solutions existed, their difference z would satisfy $Kz = 0$. Choose a component whose absolute value is largest, say $|z_i| = L$. Its equation gives $4L \le 2L$, since there are two neighboring components, each of absolute value at most L. Thus $L = 0$ and $z = 0$. There cannot be two distinct solutions. Existence for every right-hand side b can be verified directly by substitution in the explicit solution formula

$$T = \frac{1}{24} \begin{bmatrix} 7 & 2 & 1 & 2 \\ 2 & 7 & 2 & 1 \\ 1 & 2 & 7 & 2 \\ 2 & 1 & 2 & 7 \end{bmatrix} b.$$

Each b_i is a sum of two boundary values, so both stages of the construction use only fixed linear combinations.

Conclusion. The interior-temperature map is linear. Its additivity explains part (a), and its homogeneity explains part (b).